

MUNUKUTLA SRI SHANMUKHA ADITYA

Kakinada, Andhra Pradesh, India

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Professional Summary

B.Tech (Hons.) student in Data Science & Machine Learning at the Central University of Jharkhand with experience in backend development, machine learning, and AI-powered applications through research, internships, and personal projects. Interested in building intelligent software systems with a strong focus on scalable architecture, maintainability, and real-world problem solving.

Education

Central University of Jharkhand

B.Tech (Hons.) in Data Science & Machine Learning

– CGPA: 8.82 / 10

Ranchi, Jharkhand

Expected June 2027

Experience

AI Engineer Intern

Srikari Impetus Solutions Pvt. Ltd.

May 2026 – Jun 2026

Hyderabad, India

- Improved an existing tomato leaf disease classification system using TensorFlow/Keras and MobileNetV2-based transfer learning.
- Prepared the training dataset through artifact cleaning, preprocessing, augmentation, and train-validation-test partitioning.
- Implemented and trained a transfer learning pipeline using Adam optimization and categorical cross-entropy loss.
- Established a reliable training workflow with EarlyStopping and ModelCheckpoint for validation-driven model selection.

Machine Learning Research Intern

Indian Institute of Technology Patna

May 2025 – Jul 2025

Patna, India

- Implemented a supervised anomaly detection pipeline for NASA SMAP spacecraft telemetry based on a published methodology using Python and scikit-learn.
- Constructed the supervised dataset through anomaly interval mapping, preprocessing, feature engineering, and feature standardization.
- Evaluated One-Class SVM, Random Forest, and SVC models, optimizing the final pipeline using RandomizedSearchCV and SMOTE.
- Developed the final anomaly detection model achieving **88%** accuracy while prioritizing anomaly precision and recall under severe class imbalance.

Projects

Quokka – AI-Powered Conversational Decision Support System

Python, PyTorch, Transformers, SentenceTransformers, LoRA, FAISS, Mistral-7B, BGE

- Designed a modular conversational AI system to identify conversational phases and recommend personalized game-based interventions.
- Fine-tuned **Mistral-7B** using **LoRA** on a counselor-style conversational dataset generated through preprocessing and synthetic dialogue expansion.
- Built a phase retrieval pipeline using contrastively fine-tuned BGE embeddings and phase-level centroid retrieval to improve retrieval quality.
- Engineered a multi-stage conversation workflow with adaptive switching between the foundation model and fine-tuned adapter for progressive user-state validation.

CUJ Telugu Community Platform | Java, Spring Boot, React, PostgreSQL, Cloudinary

[GitHub](#) | [Live](#)

- Built and deployed a full-stack community platform for the Telugu community at the Central University of Jharkhand, enabling students, alumni, and faculty to connect through a unified portal.
- Implemented secure session-based authentication, role-based authorization, and administrator verification, with user profiles, community directory, event management, and gallery modules backed by PostgreSQL and Cloudinary.
- Designed a layered Spring Boot backend to improve maintainability and support future feature expansion.

Routr – Scalable URL Shortener | Java, Spring Boot, PostgreSQL, Redis, React

[GitHub](#) | [Live](#)

- Built a scalable URL shortening service to explore production backend design, including identifier generation, caching, analytics, and request optimization.
- Implemented deterministic short URL generation using **Base62** encoding of database-generated identifiers.
- Designed a **Redis** cache-aside architecture for low-latency redirection and buffered analytics with scheduled PostgreSQL synchronization.
- Evaluated architectural trade-offs including eventual consistency, scheduler throughput, and transient data loss, proposing event-driven synchronization and rate limiting as future enhancements.

Technical Skills

Languages: Java, Python, C, C++, SQL

Frameworks: Spring Boot, React, PyTorch, Scikit-learn, Hugging Face, SentenceTransformers

Databases: PostgreSQL, MySQL, Redis, FAISS

Tools: Git, Docker, Bash, Maven, VS Code, Postman

Core Concepts: REST APIs, OOP, DSA, Machine Learning, Deep Learning, RAG, LLMs